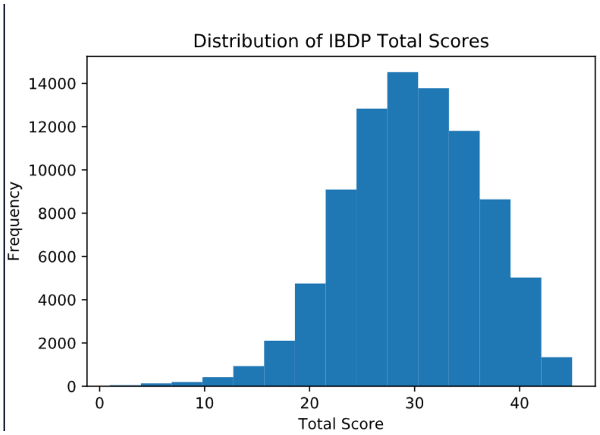
Data Science with Python
Ticket Out the Door
Set 35 Ticket Out The Door

Name ____ Period _____

Skill 35.1 Exercise 1
The International Baccalaureate Degree Programme (IBDP) is a world-wide educational program. Each year, students in participating schools can take a standardized assessment to earn their degree. Total scores on this assessment range from 1-45. Based on data provided here, the average total score for all students who took this exam in May 2020 was around 29.92. The distribution of scores looks something like this,  Statistics Academy offers a 5-week test-preparation program. A random sample of 100 students from the online prep program had an average score of 31.16. Does this sample result automatically prove that the prep program improved scores? Explain why or why not.

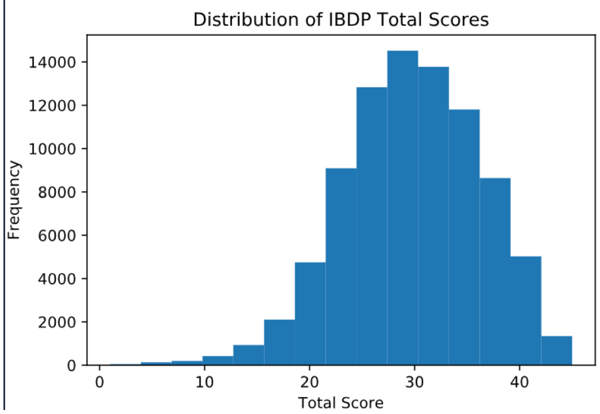
Skill 35.2 Exercise 1
Refer to the previous exercise. The average IBDP score for all students was 29.92. A random sample of 100 students from the online prep program had an average score of 31.16.
Write the null hypothesis.
Write the alternative hypothesis.
State the hypotheses symbolically and explain what each hypothesis is saying about the mean score of students in the online prep program.

Data Science with Python
Ticket Out the Door
Set 35 Ticket Out The Door

Name ____ Period _____

Skill 35.3 Exercise 1

The average total score for all students who took the IBDP exam in May 2020 was around 29.92. The distribution of scores looks something like this,



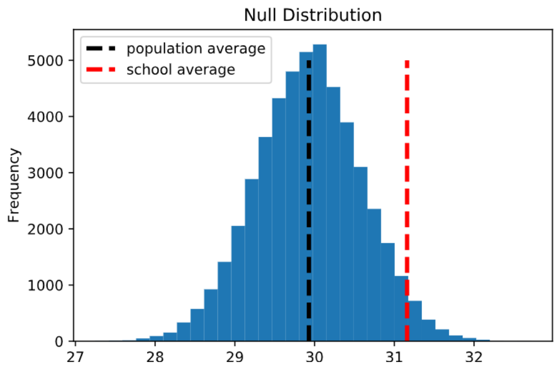
Use *np.random.choice()* to generate a null distribution of sample means for samples of size 100. Do this by calculating the mean of 500 random samples from *student_population* without replacement.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Use the *sns.histplot()* method to create a histogram of the distribution.

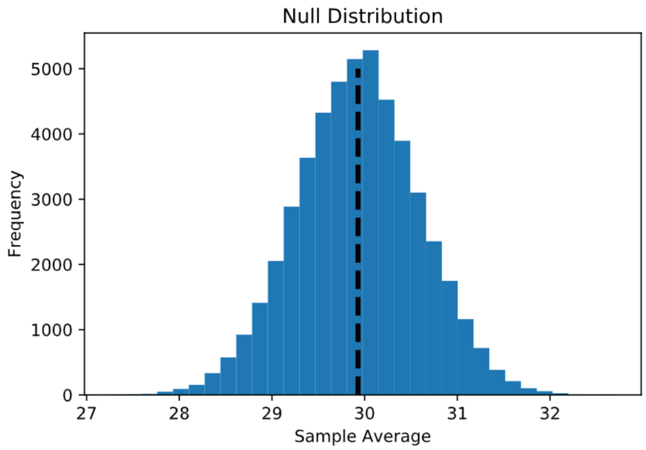
Data Science with Python
Ticket Out the Door
Set 35 Ticket Out The Door

Name ____ Period _____

The actual plot of the null distribution is shown below.  Does the school average appear common or unusual if the null hypothesis is true?
What does this suggest about the claim that the prep program improved scores?

Data Science with Python
Ticket Out the Door
Set 35 Ticket Out The Door

Name ____ Period ______

Skill 35.4 Exercise 1
The average total score for all students who took the IBDP exam in May 2020 was around 29.92. The null distribution plot is shown below,  For each situation: 1. Identify whether the test is left-tailed, right-tailed, or two-tailed 2. shade the appropriate region of the null distribution on the graph ○ A sample mean of 31.16 is used to test whether test scores are higher than average. ○ A sample mean of 29.00 is used to test whether test scores are lower than average. ○ A sample mean of 31.16 is used to test whether test scores are different from average.

Data Science with Python
Ticket Out the Door
Set 35 Ticket Out The Door

Name ____ Period ______

Skill 35.5 Exercise 1

The code below loads the population of all student scores on the IBDP exam and calculates the population mean. A sample of 100 students from Statistics Academy is stored in a list called *academy_sample*.

Use the *ttest_1samp()* function to calculate and print the p-value for:

- a **right-tailed** test
- a **two-tailed** test

for the *academy_sample*.

```
data = pd.read_csv("IDBP_scores.csv", delimiter="\t")
pop_scores = data.scores
pop_mean = scores.mean()
academy_sample = [
    18.7, 22.4, 24.1, 25.6, 26.3, 27.5, 28.1, 28.9, 29.4, 29.8,
    30.1, 30.4, 30.7, 31.0, 31.2, 31.5, 31.7, 32.0, 32.2, 32.5,
    32.7, 33.0, 33.2, 33.5, 33.7, 34.0, 34.2, 34.5, 34.7, 35.0,
    20.3, 23.1, 24.8, 25.9, 26.7, 27.8, 28.4, 29.1, 29.6, 30.0,
    30.3, 30.6, 30.9, 31.1, 31.4, 31.6, 31.9, 32.1, 32.4, 32.6,
    32.9, 33.1, 33.4, 33.6, 33.9, 34.1, 34.4, 34.6, 34.9, 35.1,
    21.0, 23.8, 25.1, 26.1, 26.9, 28.0, 28.6, 29.2, 29.7, 30.2,
    30.5, 30.8, 31.0, 31.3, 31.5, 31.8, 32.0, 32.3, 32.5, 32.8,
    33.0, 33.3, 33.5, 33.8, 34.0, 34.3, 34.5, 34.8, 35.0, 35.3,
    19.6, 22.9, 24.5, 25.8, 26.5, 27.6, 28.3, 29.0, 29.5, 40.5
]
```

The actual p-values for the right-tailed and two-tailed tests are 0.031 and 0.062, respectively. Write a one-sentence interpretation of each result.

Skill 35.6 Exercise 1

The p-value for the right-tailed test is 0.031, and the p-value for the two-tailed test is 0.062. Using a significance level of 0.05, determine whether each result is statistically significant. Then state whether you would reject or fail to reject the null hypothesis in each case.